Lab 6 Code and Branch Coverage

Application: Irrigation Scheduler | Repository: L6_L7_37_irrigation_scheduler

Tests were added to measure line and branch coverage of irrigation_scheduler.py. Both suites achieved 100% coverage; all 48 tests, including the starter tests, passed.

Code coverage

12 test cases passed. All 38 executable statements were covered, with no missing lines. The tests cover irrigation decisions, water volume, invalid area, and schedule budget handling.

Lab 6 | code coverage | Recorded terminal output

```
lab6> python -m coverage run --source=irrigation_scheduler -m pytest -q CodeCoverageTest_irrigation_scheduler.py
............                                                             [100%]
12 passed in 0.19s

lab6> python -m coverage report -m --fail-under=90
Name                      Stmts   Miss  Cover   Missing
-------------------------------------------------------
irrigation_scheduler.py      38      0   100%
-------------------------------------------------------
TOTAL                        38      0   100%
```

Branch coverage

34 test cases passed. All 28 branch outcomes were covered, with no partial branches. Tests include threshold values, empty schedules, exact budgets, skipped zones, and budget exhaustion.

Lab 6 | branch coverage | Recorded terminal output

```
lab6> python -m coverage run --branch --source=irrigation_scheduler -m pytest -q BranchCoverageTest_irrigation_scheduler.py
..................................                                       [100%]
34 passed in 0.21s

lab6> python -m coverage report -m --fail-under=90
Name                      Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------
irrigation_scheduler.py      38      0     28      0   100%
-----------------------------------------------------------------------
TOTAL                        38      0     28      0   100%
```